Name: _____

Class: _____Date:_____

CHAPTER III REVIEW

A. MULTIPLE-CHOICE QUESTIONS

Choose the **correct** statement.

1. Compare the three numbers 0; 5 and −9.

(A) $0 < -9 < 5$; (B) $-9 < 0 < 5$;

(C) $0 < 5 < -9$; (D) $5 < -9 < 0$.

Answer: _____

2. Given the set $B = \{x \in \mathbb{Z} \mid -12 < x \leq 8\}$.

(A) $-12 \in B$ and $8 \in B$; (B) $-12 \notin B$ and $8 \in B$;

(C) $-12 \in B$ and $8 \notin B$; (D) $-12 \notin B$ and $8 \notin B$.

Answer: _____

3. Two integers a and b have a negative sum $a + b$ and a positive product $a \cdot b$. Then:

(A) $a < 0$ and $b < 0$; (B) $a > 0$ and $b < 0$;

(C) $a < 0$ and $b > 0$; (D) $a > 0$ and $b > 0$.

Answer: _____

4. Two integers a and b have a positive sum $a + b$ and a positive product $a \cdot b$. Then:

(A) $a < 0$ and $b < 0$; (B) $a > 0$ and $b < 0$;

(C) $a < 0$ and $b > 0$; (D) $a > 0$ and $b > 0$.

Answer: _____

5. Two integers a and b have a negative product $a \cdot b$ and a positive difference $a - b$. Then:

(A) $a < 0$ and $b < 0$; (B) $a > 0$ and $b < 0$;

(C) $a < 0$ and $b > 0$; (D) $a > 0$ and $b > 0$.

Answer: _____

6. Two integers a and b have a negative product $a \cdot b$ and a negative difference $a - b$. Then:

(A) $a < 0$ and $b < 0$; (B) $a > 0$ and $b < 0$;

(C) $a < 0$ and $b > 0$; (D) $a > 0$ and $b > 0$.

Answer: _____

B. EXERCISES

3.41. The integer a has the sign "−" and the natural-number part 34. Write the number a and find the opposite of a.

_____

3.42. Arrange the numbers a, b, c, d in increasing order, given:

$a = 41 + (-36)$; $b = (-9) - 6$; $c = (-8) \cdot (-7)$ and $d = (-42) : 6$.

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Calculate the value of each expression in a smart way (Exercises 3.43 to 3.44):

3.43. a) $24 \cdot 31 - 4 \cdot 6 \cdot (-19)$; b) $36 \cdot 5 - 9 \cdot [(-29) + 13]$.

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3.44. a) $63 \cdot 48 + 24 \cdot (-9) - 15 \cdot 24$; b) $15 \cdot (37 - 19) - 15 \cdot (37 + 19)$.

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3.45. Find x, given $(46 - x) \cdot (x + 17) = 0$.

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3.46. Find the multiples of 7 that are greater than -22 and less than 22.

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3.47. Find all the common divisors of 30 and 45.

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3.48. Replace each "?" with a suitable integer in the table below so that the product of any three numbers in three neighbouring cells is always 90.

?	5	?	?	?	?	?	?	−3	?	?

3.49. Replace each "?" with a suitable integer in the table below so that the sum of any three numbers in three neighbouring cells is always 0.

?	?	−8	?	?	?	5	?	?	?	?